Akshitha Manne

Data Scientist

Hyderabad

Education

Data Science, *Innomatics Research Science* 07/2024
Coursework in Python, Statistics, Data Analysis, SQL + PowerBI, ML, DL, GenAI, NLP, Computer Vision Hyderabad, India

Bachelor of Technology-Electronics and Instrumentation, *Vignan Institute of Technology and Science* 09/2021 – 07/2025
Coursework Probability and Statistics,Signals and Systems,Digital Signal Processing,Sensors and Transducers,Industrial Instrumentation,Control Systems Hyderabad, India

Skills

PYTHON
- Python: Anaconda, Jupyter Notebook, Colab Notebook, syntax, Identifiers & Operators, Array, Numpy, Data Manipulation, Pandas

Deep Learning
- AI, ML AND DL, Biological Neuron, Tensorflow, Neural Network, Forward & Backward Propagation, Optimizers.

GEN AI
- Large Language Models, OpenAI, DALL-E, GPT-3/GPT-4, Gemini, LLaMA, LangChain

Power BI
- Power BI workflow, Visualisation, Trend Data viz, Power Queries, Power Pivot, DAX, DA Expression, Web & RLS, Visual Interactions, Drill Through.

Natural Language processing
- NLTK, spacy, BOW, TF-IDF, Glove, Fast text, RNNs, LSTMs, Auto Encoders, BERT

SQL
- MYSQL, NOSQL, CRUD operation, RDBMS, Data Exploration and Data Filtering, DQL and Operators, Clauses, Joins, ACID, COMMIT, ROLLBACK.

CNN & Computer Vision
Open CV, Convolutional, TransferLearing, Object Detection, YOLO.

Advanced Statistics
- Descriptive Statistics, Probability Distribution, Data Gathering Techniques, Inferential Statistics.

Machine Learning
- Supervised, Unsupervised ML Algorithms, Naive Bayes, K-NN, Hyperparameters, Decision Tree, Linear regression, Evaluation Metrics, Regression Techniques, Regularization Techniques,Logistic Regression,Xgboost.

Projects

Conversational AI Chatbot using LangChain & OpenAI
- Developed a conversational AI chatbot using **LangChain**, **OpenAI GPT**, and **ConversationBufferMemory** for context-aware responses.
- Implemented **prompt engineering**, **conversation memory**, and **LLMChain** to enable multi-turn interactions.
- Integrated **ChatOpenAI API** with custom prompts for dynamic AI conversations.
- Built an interactive Python-based chatbot supporting continuous user queries until exit.

Consumer Complaint Classification and Ticket Routing System using NLP
- Developed an **NLP-based Ticket Routing System** to classify **164K+ consumer complaints** into **5 financial service categories**.
- Implemented **text preprocessing, tokenization, sequence padding, and embedding** using a **10K-word vocabulary**.
- Built and optimized an **ANN model** with **Embedding, GlobalAveragePooling1D, Dense, and Dropout** for multi-class classification.
- Deployed the solution using **Streamlit** for real-time complaint classification and automated ticket routing.

Fake News Detection using Recurrent Neural Network
- Developed an end-to-end **Fake News Detection System** using **RNN and NLP**, achieving **85% classification accuracy** on news article data.
- Preprocessed and analyzed **40K+ news articles** through text cleaning, tokenization, sequence generation, and feature engineering.
- Implemented **Tokenizer, Padding, Embedding, and RNN layers** using TensorFlow/Keras to capture contextual patterns in textual data.
- Deployed the model using **Streamlit**, enabling **real-time predictions** through an interactive web application.

Healthcare Cost & Risk Assessment System | Deep Learning (ANN)
- Designed and trained ANN models using **100K patient insurance records** and **50+ healthcare features** to predict annual medical expenses and classify risk categories.
- Implemented data preprocessing, feature engineering, correlation analysis, and model evaluation using **TensorFlow/Keras, Pandas, and Scikit-learn**.
- Developed a **Streamlit dashboard** enabling real-time healthcare cost forecasting and risk assessment.

Zomato AI Restaurant Assistant using LangChain & OpenAI
Developed an **AI-powered restaurant assistant** using **LangChain and OpenAI GPT-4o Mini** to answer Zomato-related user queries.

Certificates

Python | EDA | SQL | Power BI | Machine Learning | Deep Learning | GenAI at Innomatics Research Labs